Documentation and publication of 3D outcomes: Methods and Tools for Hypothetical 3D Heritage Reconstructions

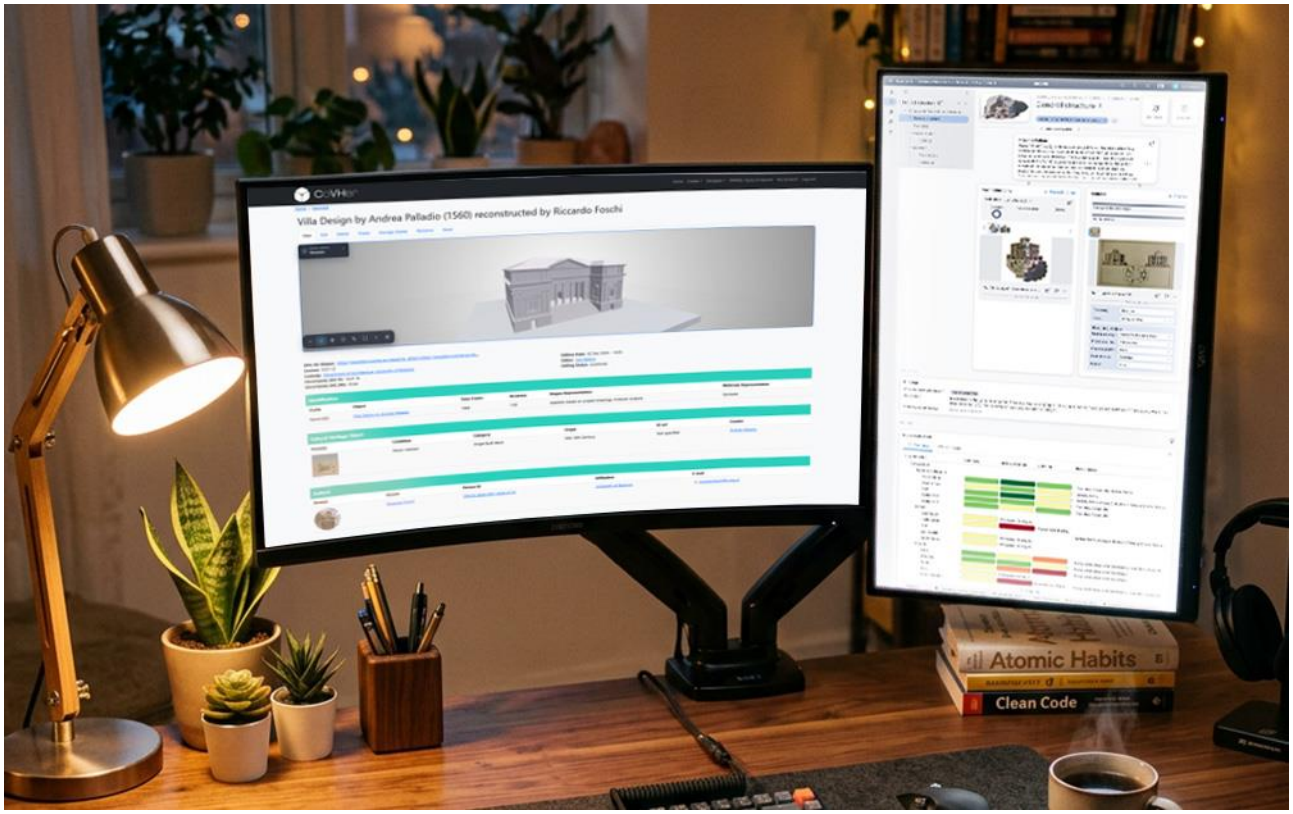


Figure: Publication of a 3D reconstruction model in the CoVHer repository, alongside the documentation of research argumentation and evaluation of results in IDOVIR.

The workshop offers hands-on experience with the documentation and publication workflow using two platforms developed for hypothetical virtual 3D reconstructions: the Infrastructure for documentation of virtual reconstructions (DFG-funded project IDOVIR) , developed within a DFG-funded project, and the repository developed as part of the Erasmus+ CoVHer project (Computer-based Visualisation of Architectural Cultural Heritage).

The first part focuses on documenting digital reconstructions based on the **Reconstruction Argumentation Method (RAM)** using the IDOVIR platform. This method uses a hierarchical subdivision of the reconstructed object and its possible reconstruction variants as the basis for documentation. The system enables source materials to be assigned to individual entities within this hierarchy, together with the corresponding reconstruction results and descriptions of the decision-making process. This approach allows participants to assess the level of uncertainty associated with their reconstruction hypotheses using predefined criteria, as well as to automatically generate structured documentation in PDF format.

The second part focuses on publishing digital models in an open 3D repository based on the WissKI Virtual Research Environment and supported by a knowledge graph. The repository aims to increase the availability of digital 3D resources relating to cultural heritage, with particular emphasis on enabling data reuse and ensuring long-term accessibility. The workshop will address the requirements for preparing 3D models for web publication, including their correct visualisation in a web browser, as well as considerations for making the published data reusable in accordance with the **Scientific Reference Model (SRM)**. The 3D model will also be contextualised through cross-references to related data and platforms, including IDOVIR.

During the workshop, participants will learn how to register on both platforms, structure their data, and prepare datasets for upload, enabling them to create accessible and transparent research records documenting both their 3D reconstruction outcomes and the research processes behind them.

The initiative is aimed at researchers, professionals, and students interested in digital reconstruction, heritage communication, and the development of transparent and shared methodologies for virtual models.

The goal of the workshop is to raise awareness of the importance of adopting shared best practices in the field of hypothetical heritage reconstructions, in order to ensure transparency, reproducibility, and verifiability of results, enabling the validation/falsification and reuse of digital artefacts produced by third parties. These principles align with the FAIR guidelines, which are more relevant than ever in light of the increasing accessibility of artificial intelligence tools capable of rapidly generating photorealistic results often without documentation. In this context, the risk of interpretative ambiguity and misinformation increases, as visual quality may lead users to perceive as fully reliable reconstructions that are, in fact, hypothetical. Therefore, the adoption of transparent communication strategies that clearly expose the decision-making process—inevitably based on inference and hypothesis—is more urgent and relevant than ever.

To register for the workshop, please send an email to: covher.workshop@gmail.com, indicating the workshop title and your full name.